

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 06



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Data transfer





Topics to be Covered



Topic

TCP Data transfer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



TCP Server Process

P_2

SOCK_STREAM
(TCP Connection
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten = P_2 's ISN

LastByteSent = P_2 's ISN

LastByteAcked = P_2 's ISN

LastByteRead = P_1 's ISN

NextByteExpected = P_1 's ISN + 1

LastByteRcvd = P_1 's ISN



Topic : TCP



- The process P_2 read N bytes from 'Recv Buffer'
- $\text{LastByteRead} = \text{LastByteRead} + N$;
- Remove N bytes from P_2 'Recv Buffer' from begin



Topic : Acknowledgment Number



=> if ACK flag is on
then TCP segment carry acknowledgment number
else :
Acknowledgment number is garbage value

=> TCP uses cumulative acknowledgment
[ACK Number = NextByteExpected ;]

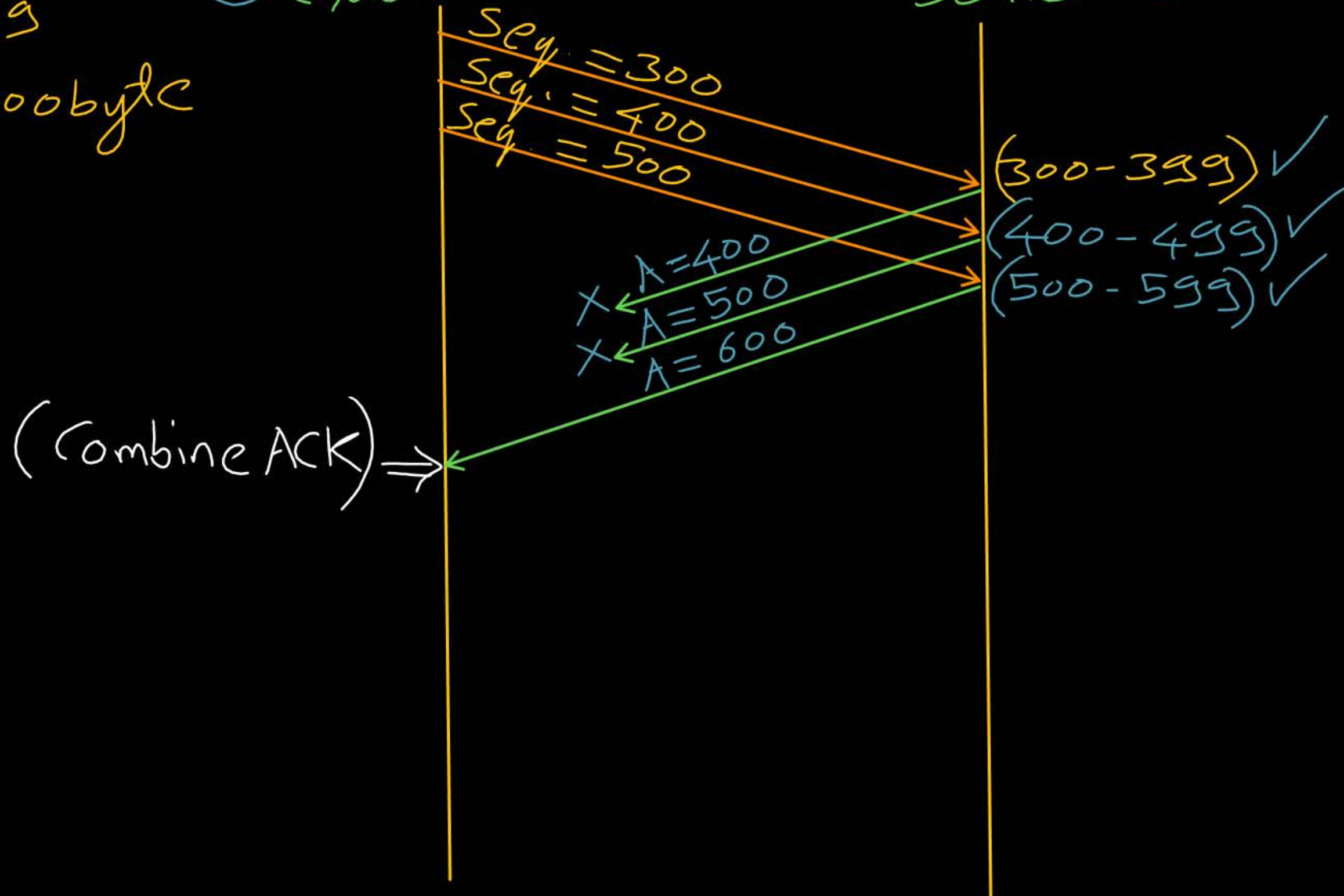


client- $ISN = 299$

$MSS = 100 \text{ byte}$

(P_1) TCP client

TCP server (P_2)





Topic : Acknowledgment Number



=> In TCP, provision for 'selective acknowledgment' (SACK)
[TCP keeps 'selective acknowledgment(s)' in optional header]

=> TCP allows for piggy-backing ✓
[sending data and acknowledgments in the same segment]



Topic : TCP

Client_ISN = 299

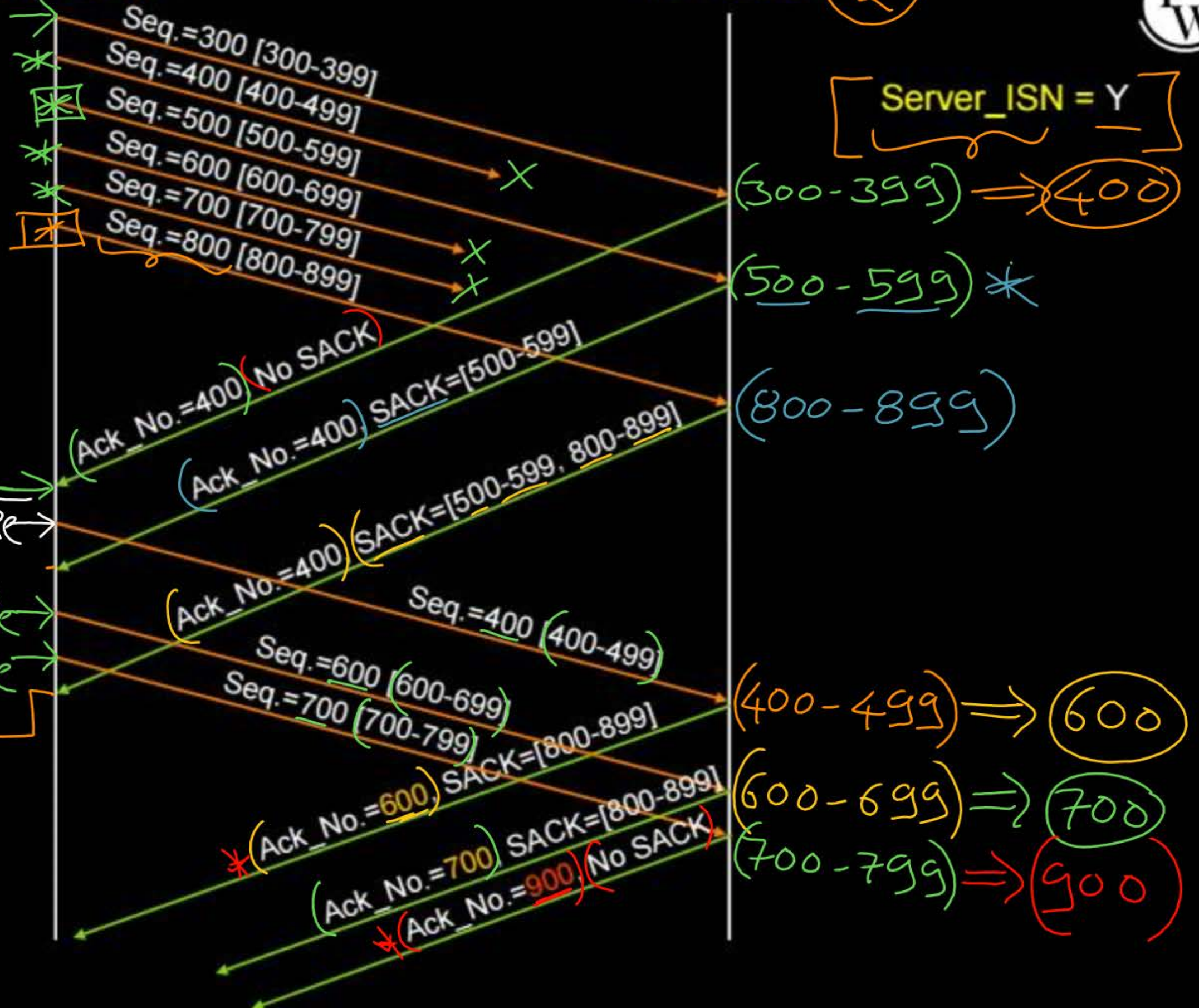
MSS = 100 bytes

P₁ TCP Client

TCP Server P₂



Server_ISN = Y



Till byte 399, Remove
from Sendbuffer

To (400) Re

To (600) Re

To (700) Re



#Q. Consider the following statements.

I. TCP connections are full duplex. TRUE

II. TCP has no option for selective acknowledgment FALSE

III. TCP connections are message streams. FALSE

(byte stream)

[GATE-2015, Set-3, 1-Mark]

IIT-K

✓ (A) Only I is correct

(B) Only I and II are correct

(C) Only II and III are correct

(D) All of I, II and III are correct

Ans. A



Topic : TCP

~~Client_ISN = 299~~

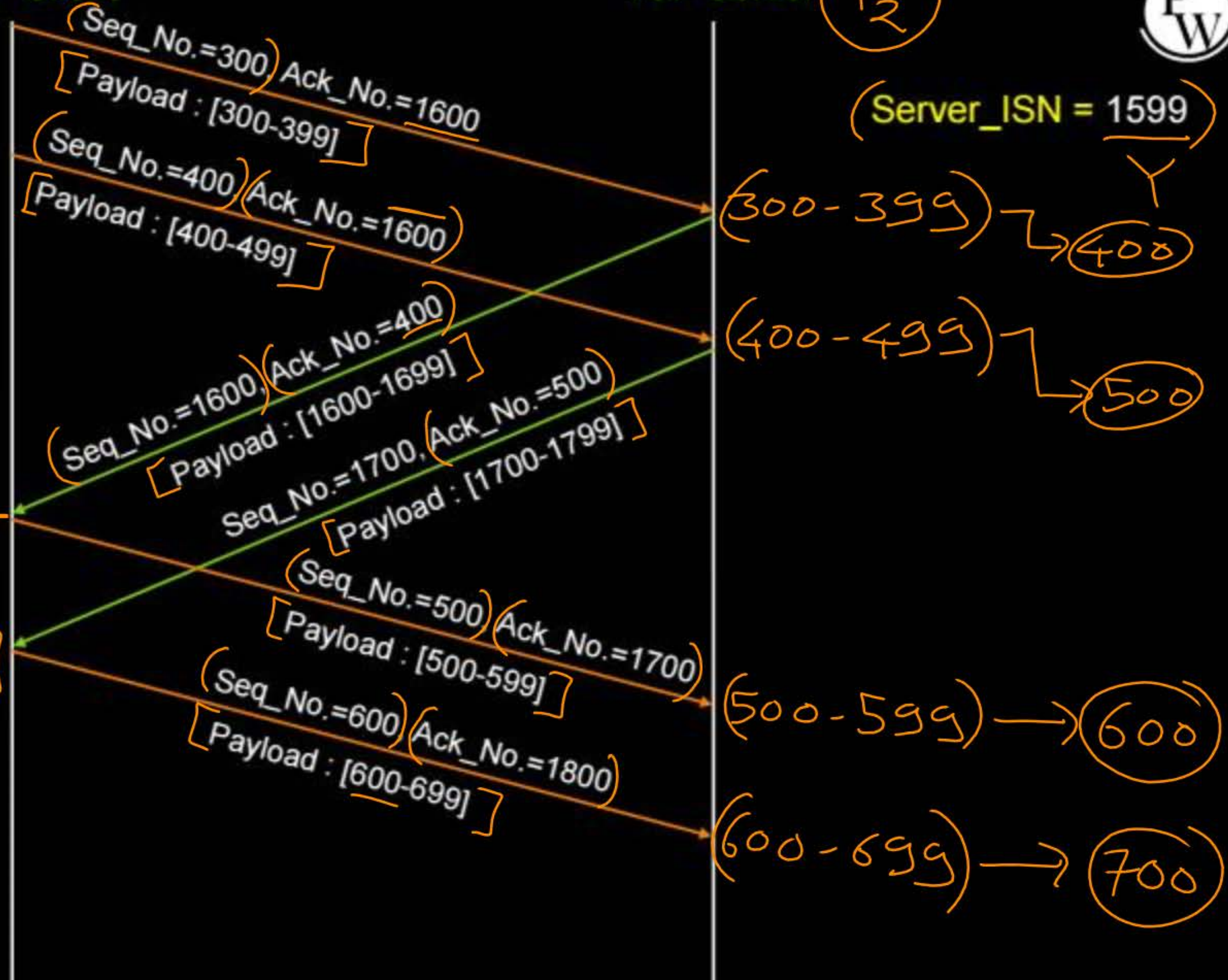
MSS = 100 bytes

P₁ TCP Client

TCP Server P₂



(Server_ISN = 1599)





#Q. Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. The values of X and Y (in that order) are :

[GATE-2007]
IIT-K, H.W.

- (A) 60 and 290
- (B) 230 and 291
- (C) 60 and 231
- (D) 60 and 230

[MCQ]

(IIT-R)

[GATE-2025] [1 Mark]



#Q. Consider the following statements :

(i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. TRUE

(ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S. FALSE

Which ONE of the following is CORRECT?

☒ A Both (i) and (ii) are TRUE

☒ B (i) is TRUE and (ii) is FALSE

☒ C (i) is FALSE and (ii) is TRUE

☐ D Both (i) and (ii) are FALSE

Ans: B



Topic : Maximum Segment Lifetime

=> Maximum Segment Lifetime (MSL)

-> Duration for a TCP segment can exist in an inter-network system

=> Wrap-around time

-> Minimum time before the sequence number can be used again



$$[MSL = \text{Wrap-around Time}]$$

P_1 SOCK_STREAM
(TCP Socket)

Send Buffer

300

300 $(2^{32} - 1)$ 0, 1, 2, 299, 300 2^{32} numbers 2^{32} bytes = 4GB

Wrap-around



Topic : Maximum Segment Lifetime



→ Suppose size of sequence number field is k bits in TCP header,
(generally k = 32 bits)

$$\text{MSL} = \frac{[2^k \text{ Bytes}]}{[\text{Bandwidth}]} = \frac{[2^{32} \text{ Bytes}]}{[\text{Bandwidth}]}$$

#Q. Consider a long-lived TCP session with an end-to-end bandwidth of $[1 \text{ Gbps}] (= 10^9 \text{ bits-per-second})$. The session starts with a sequence number of $[1234]$. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is ____.



[GATE-2018]

IIT-G

1234 $(2^{32}-1), 0, 1, 2, \dots, 1233, 1234$

2^{32} numbers $\rightarrow 2^{32}$ bytes

Wrap-around

Solution :-

$$\begin{aligned}
 \text{Wrap-around time (MSL)} &= \frac{2^{32} \text{ Bytes}}{\text{Bandwidth}} = \frac{2^{32} \text{ Bytes}}{1 \text{ Gbps}} \\
 &= \frac{2^{35} \text{ bits}}{10^9 \text{ bits/sec}} \\
 &= 34.35 \text{ sec} \\
 &= 34 \text{ sec}
 \end{aligned}$$

Ans = 34



Topic : Maximum Segment Lifetime



$$\text{Maximum number of bytes can be transmitted in one MSL} = \text{MSL} * \text{Bandwidth}$$

Minimum number of bits required for sequence number field

$$= \lceil \log_2(\text{Max no. of bytes can be transmitted in one MSL}) \rceil \text{ bits}$$

[NAT]

IIT-KGP [GATE-2022][2 Mark]



#Q. Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.

Ans = 33

Solution :-

$$\begin{aligned}
 \text{Maximum number of bytes can be transmitted in one MSL} &= \text{MSL} * \text{Bandwidth} \\
 &= 60 \text{ sec} * 1 \text{ Gbps} = 60 \text{ sec} * 10^9 \text{ bits/sec} = 6 * 10^{10} \text{ bits} \\
 &= \left(\frac{6 * 10^{10}}{8} \right) \text{ bytes}
 \end{aligned}$$

Minimum number of bits required for sequence number field

$$\begin{aligned}
 &= \left\lceil \log_2(\text{Max no. of bytes can be transmitted in one MSL}) \right\rceil \text{ bits} \\
 &= \left\lceil \log_2 \left(\frac{6 * 10^{10}}{8} \right) \right\rceil \text{ bits} = \left\lceil 32.80 \right\rceil \text{ bits} = 33 \text{ bits}
 \end{aligned}$$

[MSQ]

(H.W.) (11T-K)

[GATE-2023][2 Mark]



#Q. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

- A 30 bits
- B 32 bits
- C 34 bits
- D 36 bits



2 mins Summary



Topic

TCP Data transfer



THANK - YOU

